3.6 Measures to reduce the elements at risk

The temporary or permanent reduction of the number and/or value of the elements at risk is widely practiced and particularly cost effective, especially when the number of elements at risk is small in relation to the extent of the landslide and of the affected area and when it is achieved through the sustained implementation of appropriate long-term planning measures.

Ambrozic et al. (2009) and SafeLand (2012) distinguish between:

- Decreasing the number of vulnerable elements potentially affected by a landslide, for example by:
  - o Zoning to prevent development in hazardous areas or removing existing development from hazardous areas (exclusionary zones);
  - o Traffic restrictions (reduce number of vehicles).
- Decreasing the probability that vulnerable elements will both spatially and temporally intercept ground movements, e.g. by:
  - o Moving non-stationary vulnerable elements to less hazardous locations;
  - o Increasing awareness, detection and warning of hazards (either detected movement or trigger conditions) and subsequent avoidance (evacuation or temporary exclusion, followed by inspection before resuming normal use).

Each of these strategies can be implemented forcibly through standards and legislations or, less invasively, by means of incentives or disincentives introduced through planning. These actions could be:

- Relocation of existing facilities: Existing facilities can be completely eliminated or they can be reconverted to uses which imply a lower vulnerability to landslides;
- Reduction of specific value: The average number of people and/or the value of economic activities associated with a specific element at risk can be reduced, for example by limiting the range of end uses allowed through the planning instruments;
- Avoiding the construction of new facilities: The forced relocation of existing facilities is an extremely invasive measure, potentially applicable only in the most serious situations. A more practical approach in many cases may be the implementation of a long term strategy to prevent the location of new elements within hazardous areas, either by enforcing planning limits or through policies based on incentives or disincentives